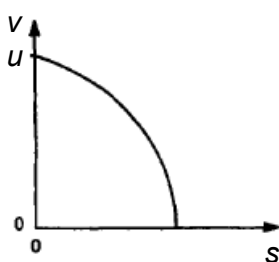
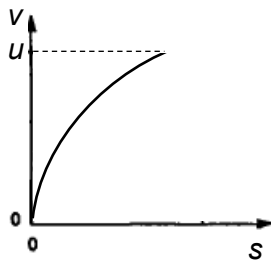


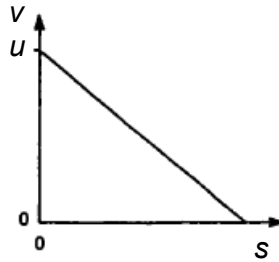
- 3 A stone falls vertically and strikes soft ground with speed u . The stone experiences constant deceleration until it comes to rest. Which graph shows the variation of speed v with distance s below the ground surface?



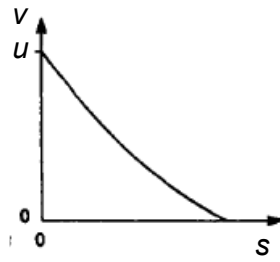
A



B



C



D

Ans: A

Take downward as positive.

Below ground surface, acceleration is upward, which is thus negative.

Let this acceleration be $-\alpha$.

$$v^2 = u^2 + 2as$$

$$v = \sqrt{u^2 + 2as}$$

$$v = \sqrt{u^2 - 2\alpha s}$$

Using GC, the graph will give be similar to option A.

- 4 A particle X with kinetic energy E collides with a stationary particle Y. Both particles have the